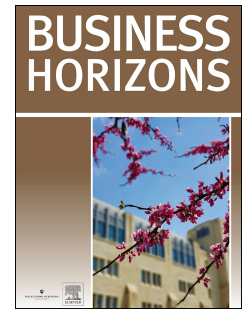


Journal Pre-proof

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Christopher Williams, Sander van Triest, Helena Kloosterman



PII: S0007-6813(25)00204-6

DOI: <https://doi.org/10.1016/j.bushor.2025.12.004>

Reference: BUSHOR 2118

To appear in: *Business Horizons*

Please cite this article as: Williams C., van Triest S. & Kloosterman H., Physicians as process innovators in healthcare organizations: Exploring the determinants of change from within, *Business Horizons*, <https://doi.org/10.1016/j.bushor.2025.12.004>.

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**Physicians as process innovators in healthcare organizations:
Exploring the determinants of change from within**

Christopher Williams ^{*}, ^a
christopher.williams@essca.fr

Sander van Triest ^b
s.p.vantriest@uva.nl

Helena Kloosterman ^b
h.kloosterman@uva.nl

^a ESSCA School of Management
55 Quai Alphonse le Gallo
92513 Boulogne-Billancourt
France

^b Amsterdam Business School
Universiteit van Amsterdam
Plantage Muidergracht 12
1018 TV Amsterdam
The Netherlands

*Corresponding author

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Abstract

This article provides evidence-based recommendations for how healthcare organizations can encourage their medical physicians to engage in process improvements and change from within. It tests the salience of three key organizational level factors that have the potential to influence professionals' process innovativeness: (1) the organization having a cost strategy that is visible to the physician, (2) the extent of human intellectual capital in the organization surrounding the physician, and (3) the sharing of common values across the healthcare organization. We draw on an online questionnaire survey among physicians in the Netherlands and use business consultants as a contrast group to understand whether physicians are different from managerially oriented professionals. We find that shared values matter for process innovativeness amongst physicians but not business consultants, while a cost strategy and abundant human intellectual capital are important for process innovativeness amongst business consultants but not physicians. We translate these findings into recommendations for physician engagement in organizational improvement and change programs in healthcare.

KEYWORDS: Process innovativeness; Healthcare; Physicians; Business consultants

1. Physicians and healthcare process innovation

Healthcare organizations are under pressure to offer cost-efficient services for patients, requiring process innovation at the front-line. Consequently, medical physicians employed by healthcare organizations are increasingly required to engage in process innovation, i.e., creating improvements and changes in their own organizational and administrative systems (Thakur et al., 2012). This requires them to exhibit process innovativeness at an individual level, i.e., to be personally involved in seeking changes in how things are done within the organization (Mylopoulos & Scardamalia, 2008). Individuals in organizations engage in process innovation when they come up with new ideas, find improved ways to do things, create better processes and routines, and work to implement new ideas (Welbourne et al., 1998). Process innovativeness is different to technological innovation because it deals with organizational innovation (new ways of organizing), administrative innovation (improvements to efficiency and effectiveness of processes and systems), and managerial innovation (changes in the way managers function) (Damanpour & Aravind, 2012).

While physicians in managerial roles in healthcare organizations can contribute to organizational change, like any manager (Kimberley & Evanisko, 1981; Savage et al., 2020), physicians that do not hold formal managerial or leadership roles within a healthcare organization also have the potential to contribute to process improvements and change. Indeed, given their dual roles in medical practice and organizational management of health service resources, all physicians can contribute to novel solutions for enhancing processes and improving cost-efficiency. Their knowledge of both organizational and patient perspectives puts physicians in an ideal position to identify opportunities for organizational improvements such as service line changes, ways of decreasing wait times, and productive use of technology and data. These changes may benefit the patient and the organization, especially when they contribute to managing costs and improving efficiency. Rising healthcare costs have been seen across all OECD countries and spending on healthcare is one of the largest areas of expenditure for governments (Papanicolas et al., 2018). Process innovativeness amongst physicians on the frontline of healthcare delivery is important to understand if healthcare organizations are to manage these cost pressures (Cadeddu et al., 2023; Lemon, 2018; Thakur et al., 2012). In the literature on physicians-as-managers, such process innovativeness is seen as a problem-solving ability in an organizational sense, rather than a medical sense (Buchanan et al., 1997; Malby et al., 2011), putting an emphasis on the organizational environment in which the physician works. Frontline actors such as physicians are crucial to the transformation of care through this ‘employee-driven-innovation’, but there is still little understanding of which factors encourage this (Cadeddu et al., 2023; Thakur et al., 2012), including factors at organizational level that healthcare administrators have some control over. Our guiding question is: how do organizational level factors influence the process innovativeness of physicians in practice?

We develop and test hypotheses relating to three commonly discussed organizational factors that are assumed to promote process innovativeness in employees. These are: the presence of a cost strategy in the organization, an abundance of human intellectual capital, and the sharing of values across the organization. Prior literature does not clearly show how these three factors apply to physicians and their engagement in process innovation in a healthcare setting (e.g., Cadeddu et al., 2023; Thakur et al., 2012). To understand whether physicians approach process innovation in a similar way to other managerial professionals, we compare physicians with business consultants in terms of how these three organizational level factors

influence process innovativeness. Physicians are increasingly expected to think and act with an administrative and bureaucratic mindset (Penberthy & Penberthy, 2020), this having roots in the growth of market mechanisms in healthcare delivery (Scott, 2004). However, as Bååthe and Norbäck (2013) note, physicians are trained differently, and have a different mindset and identity from administrators, making it difficult to know how best to ‘manage’ physicians to engage in process innovation. Ways of achieving this are discussed in the literature, such as incentivizing physicians to engage in cost-cutting initiatives through financial incentives into their own professional areas (Vilendrer et al., 2022). We propose that a better understanding of the organizational determinants of process innovativeness in physicians can be achieved through a comparison with business consultants, these being representative of managerial professionals with a focus on process improvements. This allows a reflection and critique of whether physicians can be expected to think and act as managerial professionals as far as change improvements are concerned.

2. Organizational determinants of process innovativeness amongst physicians

The internal environment within an organization has a role to play in encouraging process innovativeness amongst employees (Damanpour & Aravind, 2012; Kimberly & Evanisko, 1981). This applies to healthcare settings as much as it does to other types of organizations (Cadeddu et al., 2023). In our study, we consider three prominent ways in which the internal organizational environment may encourage process innovativeness amongst physicians. First and foremost, professionals working in healthcare provision may be encouraged to engage in process innovation when the organization emphasizes cost-effectiveness and efficiency in its strategy (i.e., when it has a *cost strategy*). Such a cost strategy must be visible to the employee and to others; this offers legitimacy for engagement in process innovation. Secondly, they may also be spurred to improve the way things are done when colleagues in the immediate environment are skilled and possess specialist expertise (i.e., when there is a high level of *human intellectual capital*). This matters as knowledge and skills within teams and groups lie at the heart of all innovative endeavors within organizations; successful process improvement projects are rarely attributed to the capabilities of just one individual. Thirdly, the extent to which the healthcare organization emphasizes the sharing of values across the organization (i.e., *shared values*) can play a role. This serves to build common alignment to goals, and it helps to stimulate employees to look for possibilities to improve processes to increase organizational performance.

2.1. Cost strategy in the organization

A cost strategy is present when there is a strategic emphasis placed by the organization on making processes more cost-efficient, and improving the utilization of available equipment, services and facilities (Auzair & Langfield-Smith, 2005). The emphasis on improving the cost required for coordination and delivery of various services is highly relevant in all advanced healthcare systems (Bååthe & Erik Norbäck, 2013; Papanicolas et al., 2018). A healthcare organization that emphasizes a cost strategy through its communication with stakeholders, provides signals from the top leadership team concerning budgetary constraints and the need for efficiency improvements. These signals give legitimacy to physicians to engage in process innovation. Physicians will be more alert to opportunities for process improvements aimed at cost-efficiency and providing value-for-money when they have this reinforcement from top management. They will feel legitimized to take actions in pursuit of these opportunities when the healthcare organization has a cost strategy.

- Hypothesis 1: Process innovativeness amongst physicians will be associated with an espoused cost strategy by the healthcare organization.

2.2. Human intellectual capital surrounding the physician

In healthcare organizations, physicians are surrounded by experience, skills, and knowledge in the wider organization: other physicians, nurses, administrators who deal with patients, and even patient support groups and representatives. This wider collection of human intellectual capital provides a foundation for innovativeness largely due to the depth of experiential and tacit knowledge that it embodies. New knowledge is created when physicians engage in process innovation; this being done in addition to their daily practice (Mylopoulos & Scardamalia, 2008). They work with others to bring their existing knowledge to problems of routine practice (acting as ‘routine experts’), as well as seeking to improve and change practices through problem solving (acting as ‘adaptive experts’) (Mylopoulos & Scardamalia, 2008). The human intellectual capital in a healthcare organization is a reflection not only of the formal training and skills-development of other professionals surrounding the physician, but also of many years of hands-on experience dealing with medical conditions and treatments. Human intellectual capital is notoriously fluid and can diffuse across organizations within and across industries. Industry knowledge is also important for process innovativeness within a particular organization: knowing how solutions have been identified and how they have worked in other settings. Human intellectual capital within the organization supports the sharing of knowledge within the organization: knowledge of opportunities and solutions for improvements in how to do things and how to extract cost savings and generate improved outcomes for patients.

- Hypothesis 2: Process innovativeness amongst physicians will be associated with human intellectual capital within the healthcare organization.

2.3. Sharing of values in the organization

It is important that all employees – including physicians - align with the goals of the organization if they are to pursue innovative behaviors on behalf of the organization. This will help ensure that their actions will be directed in the interests of the organization, rather than in individual self-interest. Shared values means that personal values and goals are consistent with those of the organization, and that these values are promulgated throughout the organization. Given that medicine is a strong professional field, any inconsistency or incongruity between the values of the organization (as the employer of the physician) and the values of the profession (to which the physician belongs) can lead to organization-professional conflict (Tummers et al., 2012), with negative outcomes such as an increase in the likelihood of employee turnover. The sharing of values is seen as an informal form of control, different to formal mechanisms such as performance appraisal and monitoring. It aims to establish trust between the employee and the wider organization, while it also ensures that employees appreciate and value the organizational goals. This is particularly important in healthcare where communities of practice are formed for complex problem solving, these being increasingly recognized as inter-disciplinary in nature (Boon et al., 2019). Shared values are important for process innovation: they motivate employees to identify improvement opportunities to increase organizational performance, while the increased level of trust will help alleviate uncertainties and concerns over creativity and risk-taking, especially as far as change improvement programs are concerned. Establishing practices for

frontline idea generation and participation in innovative activity is supported when there are shared values within the organization.

- Hypothesis 3: Process innovativeness amongst physicians will be associated with the sharing of values across the healthcare organization.

3. Physicians as business consultants?

Physicians have increasingly entered the managerial sphere and are encouraged to think and act with a managerial mindset (Bååthe & Erik Norbäck, 2013; Scott, 2004; Vilendrer et al., 2022). This is reflected in core training programs (Busari, 2012) and the fact that some physicians decide to take on MBA programs as part of a dual-degree basis for their careers (Lemon, 2018). Laditi et al. (2023) report that the “percentage of schools offering MD/MBA programs increased from 26.4% in 2002 to 60.9% in 2022” although less than 1% of medical students graduate with an MD/MBA. Ljuboja et al. (2016) note that most physician-MBAs subsequently divert their careers away from clinical activity following their education. Even those without an MBA can move permanently or semi-permanently into administration roles (Lee & Hoyle, 2002). It is in the managerial realm that they become sensitized to tools and techniques for assessing cost-efficiency and process improvements, and where they will learn and apply approaches in change management and organizational transformation.

Some scholars have observed conceptual similarities between the capabilities and practices of physicians and business consultants. Parallels have been made with respect to diagnosis of a reported issue, selecting a treatment or intervention path, and monitoring progress over the longer-term (Evers & Menkhoff, 2004; Lehene, 2021; Zimmel, 2018). The physician-patient relationship in medical practice and the consultant-client relationship in business consultancy both involve a professional applying knowledge and skill to a diagnosed problem or condition, in an environment of trust and confidence. Both are professions in which ethical standards should be followed, and where there has been ethical misconduct in the past. And both physicians and business consultants have come under increasing pressure to contribute to improvements and changes in their own organizational systems, i.e., to be innovative within the organization, for the organization (Busari, 2012; Williams, 2019).

However, there are areas of difference. One area of difference is that process innovativeness amongst physicians is not seen as a primary focus - medicine and healthcare delivery are - while in business consultancy process innovativeness is assumed to be a primary capability of professionals within the field (Williams, 2019). Physicians’ innovativeness in the organizational sense (not the clinical sense) has not traditionally been an expected norm, and the role is not without critics. Shaw (2020) notes that physicians’ patients are in a greater position of vulnerability than business consultants’ clients. He also emphasizes how the physician-patient relationship does not purely have a contractual, commercial basis in the same way that a business consultant-client relationship does. The medical profession has a social contract with the community it serves (Lee & Hoyle, 2002).

These areas of commonality and difference are rooted in the nature of the respective professional fields. Medicine is seen as a strong professional field, with strict training and regulative frameworks regarding ethics, standards, and the development and application of knowledge. Business consultancy is seen as a weaker professional field, with less emphasis

on formal training and regulation, and consequently more room for innovative behavior. Consequently, identity clashes between physicians and the managerial realm have been reported (Bååthe & Norbäck, 2013) while there is increasing pressure on physicians to think and act as business consultants in the bureaucratic management of their profession (Busari, 2012; Lee & Hoyle, 2002; Lemon, 2018). Table 1 provides a summary of similarities and differences between physician and business consultancy fields as far as process innovativeness is concerned.

[INSERT TABLE 1 HERE]

Despite these stylized similarities and differences, it is acknowledged that the managerial and organizational system surrounding a physician can influence their propensity to engage in process innovation (Bååthe & Erik Norbäck, 2013; Vilendrer et al., 2022). We argue that comparing the determinants of process innovativeness between physicians and business consultants is useful if we are to understand whether it is even possible for physicians to think and act more like business consultants as far as process innovation is concerned. Understanding whether physicians are stimulated to engage in process innovation in the same way as business consultants is important but has not been achieved in prior research.

4. Fieldwork and results

4.1. Method

The empirical fieldwork involved an online questionnaire survey of professionals that allowed us to test the three central hypotheses on our main target group, i.e., physicians, as well as on a comparison group of business consultants. The study took place in The Netherlands, an economically developed country with one of the most advanced healthcare systems in the world, ranking 11th out of 167 countries in 2023 (Statistica, 2023). While healthcare systems differ structurally across countries, it has been noted that professional identities in medicine are more coherent across countries (Bååthe and Norbäck, 2013), reinforcing the potential generalizability of our findings. Nevertheless, future research in other countries should be conducted as a replication exercise to examine the extent to which the findings reported here are translatable to other healthcare systems. We used a networking approach to identify and contact respondents. This was a “respondent-driven” (“snowball”) sampling approach (Salganik & Heckathorn, 2004), which is suitable for populations where a sampling frame is not available. A cover letter and ethics statement were given to respondents, as well as a qualification question requiring them to have at least three years’ professional work experience.

The dependent variable – *process innovativeness* – was constructed as a scale from four items on the questionnaire that asked each respondent to rate their own performance in terms of (1) coming up with new ideas, (2) working to implement new ideas, (3) finding improved ways to do things, (4) creating better processes and routines for the organization (Welbourne et al., 1998). The first explanatory variable – *cost strategy* – was constructed from 4 items relating to the emphasis placed by the healthcare organization on (1) making services/processes more cost efficient, (2) improving the utilization of available equipment, services and facilities, (3) achieving lower cost of services than competitors, and (4) improving the cost required for coordination of various services (Auzair & Langfield-Smith (2005). The next explanatory

variable – *human intellectual capital* – was constructed from five items relating to the extent to which the respondent felt they were surrounded by colleagues who were (1) highly skilled, (2) widely considered the best in the industry, (3) creative and bright, (4) experts in their particular jobs and functions, and (5) able to develop new ideas and knowledge (Subramaniam & Youndt, 2005). The final explanatory variable – *shared values* – was constructed from four items relating to the extent to which the respondent felt they experienced opportunities for (1) establishing similar personal values with the organization, (2) attending regular social events hosted by the organization, including team building, sporting events, fundraisers and volunteer events, (3) creating a sense of “ownership” for the organization, (4) understanding organizational core values (O’Reilly & Chatman, 1986; Thakur et al., 2012). We checked the internal consistency of these scales, i.e., that items on each scale represent the same underlying construct, using Cronbach’s alpha. Alpha can take a value of between 0 and 1, and values over 0.7 are seen as acceptable, while values over 0.9 may be a sign of internal redundancy with too many items testing the same question (Tavakol & Dennick, 2011). In our data, *process innovativeness* has an alpha = 0.75, *cost strategy* an alpha = 0.70, *human intellectual capital* an alpha = 0.85, and *shared values* an alpha = 0.80. We interpret the scale construction as reliable.

We used four control variables in the statistical tests, variables that have the potential to explain process innovativeness amongst physicians. Firstly, we controlled for the experience of the individual physician in their professional field, captured in years. This can be important because of knowledge gained over time on how to solve process problems and how solutions in different settings have taken effect. Research supports this; for instance, hospital administrator tenure is positively linked to adoption of technological innovations in hospitals (Kimberly & Evanisko, 1981). Experience in the wider professional field will improve judgement and contribute to the overall memory of the organization (Tien et al., 2019), useful when seeking new ways of working and developing new routines. Secondly, we captured self-reported gender as this can be associated with innovative behaviors. Female leaders tend to have a more transformational leadership style and are also linked with innovation-supportive organizational cultures (Wu et al., 2021). In the nursing profession, Nählinder (2010) shows females have higher levels of creativity and innovativeness when measured by the number of new ideas that are related to solving problems in an everyday healthcare setting. Thirdly, we controlled for the complexity of the task performed by the physician as this can influence their desire to change the way things are done. At the heart of task complexity is whether non-routine problems occur for the individual in their daily work, i.e., problems for which new solutions are needed (Van de Ven & Ferry, 1980; Glover et al., 2020). We used an item capturing whether the respondent felt they encountered problems in their daily work for which there were no immediate solutions. Fourthly, we accounted for organizational size; this being potentially linked to burdensome bureaucracy and resistance to change.

4.2. Findings

We used factor analysis to validate the scales and correlation and regression analysis to assess the relationship between the organizational level variables and process innovativeness. The final sample sizes after removing observations with missing values were $n=61$ for physicians and $n=120$ for business consultants. A t-test on the dependent variable showed no statistical difference between the two groups, that is, on average they report similar levels of process innovativeness. Information criterion tests showed that both the physician model and the

business consultant model fit well. We also ran the regressions as Bayesian tests which are suitable for small sample sizes, and the pattern of coefficients is consistent with the main results reported below. The results of the correlations are shown in Table 2, and the regression analysis is shown in Table 3.

[INSERT TABLES 2 & 3 HERE]

The correlation comparison shows key differences in the coefficient r between the two samples. This coefficient is an indication of the strength and direction of the association between two numeric variables. It ranges from -1 (a perfect negative correlation) to +1 (a perfect positive correlation). A value of 0 indicates no association whatsoever. Cost strategy and shared values are associated in healthcare organizations ($r=0.37$), but not in the consultancy firms ($r=0.07$), while organizational size and cost strategy are negatively associated in healthcare organizations ($r=-0.30$), but positively associated – albeit weakly - in the consultancy firms ($r=0.16$). This suggests that healthcare organizations understand that cost effectiveness is important and try to include it in their values, more so than consulting organizations. It also suggests that smaller healthcare organizations are particularly keen on managing costs.

The regression results indicate, surprisingly, that organizational cost strategy does not matter to the proclivity of physicians to engage in process innovation within their organization, providing no support for H1. However, it does have a positive effect for business consultants. The results indicate that the extent of human intellectual capital within the organization does not matter for the proclivity of physicians to engage in process innovation within their organization, providing no support for H2. Again, it does have a positive effect for business consultants. In terms of shared values, the results indicate that the extent of shared values within the organization does matter to the proclivity of physicians to engage in process innovation within their organization, providing support for H3. This factor does not have any effect for business consultants.

The results for the control variables are the same when contrasting physicians with business consultants. The main result is that it is experience in the profession that is most important to the proclivity of both physicians and business consultants to engage in process innovation within their organization. There was no support for gender (Nählinder, 2010) or task complexity (Van de Ven & Ferry, 1980; Glover et al., 2020).

5. Recommendations for healthcare organizations

These findings translate into several recommendations for encouraging process innovativeness amongst physicians in practice. Leaders and administrators of healthcare organizations that seek cost-efficiency through continuous process improvement can harness their existing body of physicians as a vital source of process innovativeness. External business consultants that are brought into healthcare organizations for this purpose also have a role to play, but, as we found, their triggers for process innovativeness are different to physicians and these differences are important to understand and manage, particularly if physicians and external business consultants work alongside each other.

5.1. Nurture shared values across the healthcare organization

The result for the effect of shared values on process innovativeness in physicians is striking in our study. It suggests how vitally important it is to avoid any situation in which the personal values of the physician become out of sync with those of the employer. When physicians share the values of the organization, they are willing to change their own behavior to improve organizational performance. Healthcare organization administrators can have regular informal conversations with physicians to understand share values and the fit – or lack thereof - between personal and organizational values. They can use these conversations to monitor and evaluate whether this fit is improving or worsening across the organization over time. And they can review the extent to which socialization is used within the organization to bind employees together and create a sense of shared responsibility and common purpose. Involving physicians in the development and dissemination of core values throughout the organization will help create a sense of ownership for the organization amongst physicians. Creating opportunities for socialization within the organization is part and parcel of this; this can extend to social events to which family members are invited, team building, fund-raising and charity events, and sporting events.

5.2. Leverage experienced physicians for process innovation

The strong effect of professional experience on process innovativeness at an individual level suggests healthcare administrators need to have a clear view of prior experience of physicians. One way of doing this is to maintain a register of physicians according to experience. This can include what types of professional experiences the physician previously had in terms of sub-disciplines and fields, what specialist skills they developed, and the durations of these experiences. This information can be drawn on when forming temporary project teams for process innovation. Healthcare organizations should not, however, rely solely on the more experienced physicians for creative contributions; this may come at a cost of time spent away from vital patient care where experience is crucial. What it means is that experienced physicians can be incentivized to participate in process innovation projects at a level that makes sense, given their other commitments (Brüggen et al., 2018). Experienced physicians can have a mentoring role in process innovation projects that involve more junior colleagues, sharing their broader experience in organizational transformation and change. Junior colleagues can learn from senior physicians in a process innovation sense as well as a medical sense when co-working together.

5.3. Seek complementarity with business consultants, don't emulate them

We noted above how physicians have increasingly pursued training and career development in management and organizational leadership (Buchanan et al., 1997; Busari, 2012; Malby et al., 2011; Quinn & Perelli, 2016). By choosing to study MBA programs (Lemon, 2018) and moving into administrative roles (Lee & Hoyle, 2002) where they work more intensively with external business consultants, physicians may be tempted to emulate consultants in their thought processes and paradigms for addressing change. Our results show that physicians and business consultants are triggered towards process innovativeness by completely opposite organizational level factors. For physicians, the key factor is shared values; cost strategy and human intellectual capital do not matter. For business consultants, shared values are not important, while cost strategy and human intellectual capital in the organization matter. This underscores the importance of appreciating differences between professions (Shaw, 2020) and respecting professional identity rather than changing it (Bååthe & Norbäck, 2013). Healthcare organizations can get the best of both worlds (medical practice and business consultancy) by seeking complementarity between physicians and external consultants on

process and change management project teams. Leaders need to be aware that physicians and business consultants have different triggers for process innovativeness and reassure physicians that they do not have to emulate the behaviors or thought processes of business consultants. Business consultants will not necessarily share the same values or be sensitive to values sharing within the healthcare organization in the same way that physicians will be. Consequently, project teams combining physicians and business consultants can be established as a safe space for each side to listen to and challenge the other in the interests of the overall goal.

5.4. Choose business consultants who identify with healthcare organizational values

Healthcare organizations have processes for evaluating and selecting external consultants to work with on operational and process improvement problems. Our finding for business consultants suggests they will be stimulated to seek process improvements when they understand the nature of human intellectual capital in the organization (this having a major influence) and, to a lesser extent, when the organization espouses a cost-saving strategy. It is important that consultants understand that physicians are motivated differently to engage in improvement. If they are not aware of these factors for a specific healthcare organization, they may over-rely on ‘tried-and-tested’ solutions used in other organizations rather than making the case for improvements that support reaching the healthcare organization’s distinct objectives.

5.5. Avoid over-reliance on a top-down approach to cost strategy

While this may seem counter-intuitive for healthcare organizations that face pressures from funding sources to drive large scale cost savings top-down, our findings provide the basis for a different approach. This involves recognizing that physicians can create new ideas for optimizing processes and ensuring value-for-money and cost-efficiency without harming patients. Because of their knowledge and experience in both administrative and healthcare delivery domains, they are in an ideal position to contribute. This will involve them devoting time to creative-oriented activity in the managerial realm (process innovativeness) at the expense of routine activity (service delivery). Administrators need to accept that a trade-off is needed here and that physicians can be stimulated into engaging in creative-oriented activity for the organization through a lateral values sharing approach within the culture of the organization, and not simply a top-down approach that tends to be highly centralized. Administrators should acknowledge that physicians will play a key role in implementation and learning from process innovation projects and that they will have an important voice in how outcomes are translated in institutionalized processes through refinement and fine-tuning. This does not mean that top-down management is not important, it means that leaders need to be careful about exclusively relying on senior management directives for delivering cost-savings projects. Table 4 presents a summary of the implications for managing process innovativeness amongst physicians within healthcare organizations.

[INSERT TABLE 4 HERE]

6. Concluding remarks

We conclude that physicians do have a role to play in driving change from within in healthcare organizations. Their proclivity in doing this is influenced by the sharing of values within the healthcare organization as well as by their level of professional experience. It is

not influenced by cost strategy and human intellectual capital within the organization, or their gender, or perceived task complexity at an individual level. The reason for the lack of significance for cost strategy and human intellectual capital can be explored in future work. We suggest that the reason is deeply entrenched in the professional identities of physicians (Bååthe & Norbäck, 2013). As Bååthe & Norbäck (2013: 480) note: “the managerial agenda seldom fits the physician mindset/professional identity”, and the solution to overcoming the identity divide lies in the common sharing of values within the organization, as opposed to levels of talent in colleagues or the directives of senior leadership. This concurs with those who espouse cultural factors and social controls within organizations as ways of overcoming divisions based on different identities. Our results allow critical reflection on prior research on healthcare process innovation. Prior work creates an impression that many organizational variables will equally have a role to play in encouraging physicians to innovate on the frontline. Variables identified in the literature include: openness to innovative ideas, organizational efficiency, resource availability, management support and market orientation (e.g., Cadeddu et al., 2023; Thakur et al., 2012). Our results suggest that such variables should not be treated equally. In our study, we find that shared values is the most important factor for physicians while cost strategy and human intellectual capital may not have any role to play at all in encouraging innovativeness in physicians.

Any similarities between physicians and business consultants as far as organizational determinants of process innovativeness are concerned are difficult to establish. The organizational level factors that are associated with higher levels of process innovativeness amongst physicians are not the same as those within business consultants. This distinction is important for two reasons: (1) physicians are increasingly working within the organizational and managerial domain within healthcare and even undergo management training to think and act more like business consultants within this domain, and (2) healthcare organizations commonly employ external business consultants to work on process innovation and operational improvement programs alongside physicians. Our findings and recommendations provide a basis on which healthcare organization administrators can harness the innovative potential of physicians during important change management projects aimed at improving cost-efficiency and providing value for money, especially as they work together with external business consultants.

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Journal Pre-proof

Table 1. Stylized comparison of physicians and business consultants

Contextual feature	Context 1: Physicians	Context 2: Business Consultants	Comment
Beneficiary – who needs help?	Patient	Organization (a client)	Both physicians and business consultants aim to provide help and assistance for an identified beneficiary
Diagnosis	A defined problem is presented by a patient to a physician, with identifiable symptoms	An issue is presented by an organization to a business consultant (e.g., poor performance, change management problems), although this can be triggered by a pre-sales activity initiated by the consultant towards the client	Both physicians and business consultants perform diagnosis to identify and understand underlying reasons for symptoms
Relationship	Many physicians may treat one patient for a given symptom	One business consultant may have many contacts within a client organization during an assignment	There is a need for trust, confidentiality, and clear communication in both contexts
Cost of solution	The cost of treatment or intervention is borne by different stakeholders (state, insurance, and/or patient)	The cost of treatment or intervention is borne by the client	There is a more complex payment system in healthcare compared to consultancy
Process	Patients may be passed from primary care to secondary and tertiary specialist care, depending on diagnosis; billable utilization is not an issue as demand is high	Business consultants will seek to scope and deliver work within their capability, aiming for follow-on work for financial gain; billable hours and utilization are key considerations	Both follow a generic process of diagnosis, treatment and intervention, application of specialized knowledge, monitoring and aftercare

Contextual feature	Context 1: Physicians	Context 2: Business Consultants	Comment
Professional field	Medicine is a strong professional field, based on institutional norms, training, and legal requirements	Business consultancy is a weaker professional field	Both fields are seen as knowledge intensive; professionals within the field develop reputations over time
Process innovativeness	Physicians are increasingly asked to be involved in operational efficiency and cost savings initiatives within their organizations	Business consultants more likely to help clients with process improvements, although can work on 'in-house' change management when necessary	Both have advanced and specialist knowledge that can be brought to bear for process improvements for their employers

Table 2. Correlations and a comparison between physicians and business consultants

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Physicians (n=61)							
Process Innovativeness (1)							
Cost Strategy (2)	0.12						
Human Intellectual Capital (3)	0.18	0.04					
Shared Values (4)	0.38	0.37	0.18				
Experience (5)	0.32	0.14	0.12	0.14			
Gender (6)	0.09	0.09	-0.25	0.08	0.06		
Task Complexity (7)	0.14	-0.05	-0.01	-0.03	0.13	-0.10	
Organizational Size (8)	0.13	-0.30	0.04	-0.07	0.08	0.05	0.22
Business consultants (n=120)							
Process Innovativeness (1)							
Cost Strategy (2)	0.22						
Human Intellectual Capital (3)	0.26	0.13					
Shared Values (4)	0.14	0.07	0.32				
Experience (5)	0.20	0.17	0.10	-0.06			
Gender (6)	0.02	-0.06	-0.04	-0.01	-0.13		
Task Complexity (7)	0.10	-0.11	-0.03	-0.01	0.02	-0.08	
Organizational Size (8)	0.04	0.16	0.02	-0.06	0.01	-0.08	0.09

Table 3. Process innovativeness in physicians and a comparison with business consultants

	Physicians		Business consultants	
	(1)	(2)	(3)	(4)
<i>Organizational factors</i>				
Cost strategy		-0.01 (0.12)		0.15+ (0.08)
Human intellectual capital		0.12 (0.19)		0.18* (0.09)
Shared values		0.26* (0.11)		0.07 (0.08)
<i>Control variables</i>				
Experience	0.06* (0.02)	0.05* (0.02)	0.04* (0.02)	0.03* (0.02)
Gender	0.10 (0.17)	0.11 (0.15)	0.08 (0.12)	0.10 (0.11)
Task complexity	0.06 (0.09)	0.07 (0.07)	0.06 (0.06)	0.08 (0.05)
Organizational size	0.06 (0.10)	0.07 (0.10)	0.02 (0.06)	-0.00 (0.06)
N	61	61	120	120
Variance explained	0.13	0.26	0.05	0.15

Coefficients shown with robust standard errors in parenthesis; * indicates statistical significance at the $p < 0.05$ level; + $p < 0.1$ level

Table 4. Encouraging process innovativeness amongst physicians in healthcare organizations

Recommendation	Description	Implementation considerations
1	Nurture shared values across the healthcare organization	• Create opportunities and incentives for socialization within the organization (social events, team building, fund-raising, sporting events) • Establish core values for the organization and communicate to all employees • Create a sense of ownership for the organization amongst physicians • Have informal conversations with physicians to understand fit between personal and organizational values
2	Leverage experienced physicians for process innovation	• Maintain a register of physicians according to experience (fields, skills, durations) • Provide incentives for experienced physicians to participate in process innovation projects • Allow juniors to learn from senior physicians in process innovation projects
3	Seek complementarity with business consultants, don't emulate them	• When managing projects involving physicians and business consultants, seek complementary skills and experiences within the project team • Be aware that physicians and business consultants have different triggers for process innovativeness • Reassure physicians that they do not have to emulate the behaviors or thought processes of business consultants, and can even challenge them in the interests of the overall goal
4	Choose business consultants who identify with healthcare organizational values	• Select external business consultants to work on process innovation projects in the healthcare organization who can appreciate that physicians' process innovativeness is influenced by shared values

Recommendation	Description	Implementation considerations
5	Avoid over-reliance on a top-down approach to cost strategy	• Recognize that frontline physicians can create new ideas for optimizing value-for-money and cost-efficiency and that they are stimulated into doing so through lateral values sharing • Acknowledge that physicians will play a key role in implementation and learning from process innovation projects • Do not rely exclusively on senior management directives for cost-savings projects • Be sensitive to potential clashes and differences of opinion between administrative management and healthcare professionals on service delivery during process innovation projects